NUR E JANNAT ASHA

Software Engineer | C++, Python & High-Performance Systems

📍 **Espoo, Finland** (Helsinki Metropolitan Area) 📞 +358 417982475 ✉ jannatashanur@gmail.com

💻 github.com/Jannat332 🌐 jannat332.github.io/portfolio 💼 linkedin.com/in/nur-e-jannat-asha

✓ Full Finnish Work Rights (LUT M.Sc. Graduate) ✓ Available Immediately • Espoo / Helsinki / Remote

★ Eligible for Aalto University F.E.C. Subsidized Recruitment Traineeship English (Fluent) • Finnish (A1 – Active)

4.24 / 5.00	6 Repositories	60% Speedup	5 Publications
LUT M.SC. DISTINCTION	PUBLIC CODE & CI/CD	GPU/C++ PARALLELIZATION	IEEE & ELSEVIER JOURNALS

PROFESSIONAL SUMMARY

Software Engineer with a Master of Science in Software Engineering (**4.24/5.00 CGPA, Distinction**) from LUT University, Finland. Solid engineering track record in **C++17, Python, Linux systems, and distributed backend architectures**. Proven ability to architect deterministic state machines, accelerate computational workloads with CUDA parallelization (achieving 60% runtime reductions), and deliver production-ready software backed by automated CI/CD and rigorous test coverage. Open to full-time engineering roles or placement via the **Aalto University F.E.C. (Further Educated with Companies)** recruitment program.

TECHNICAL EXPERTISE

Core Languages:	**C++ (C++17/20, STL, RAII, Smart Pointers)**, **C**, **Python (3.10+, FastAPI, NumPy, Pandas)**, SQL (PostgreSQL, SQLite), Bash, JavaScript (React 18), CUDA
Architecture & Systems:	Object-Oriented Design (SOLID), Finite State Machines (FSM), Concurrency & Threading, Memory Safety, RESTful API Design, Microservices
DevOps, Cloud & Tooling:	**Linux/Unix (POSIX)**, **Docker** (multi-stage builds), **GitHub Actions CI/CD**, CMake, Git, GDB, Valgrind, Postman
Quality & Engineering:	Unit Testing (GoogleTest, PyTest), Test-Driven Development (TDD), Static Analysis (Cppcheck, Flake8), Code Review, Agile / Scrum
Domain Competencies:	5G Network Slicing (3GPP QoS/URLLC), LLM Verification & RAG Pipelines, Embedded Sensor Data Acquisition, Machine Control Systems
Languages:	**English** (Fluent / Full Professional), **Finnish** (A1 – actively learning in weekly courses), Bengali (Native)

FEATURED PRODUCTION REPOSITORIES

Verify code directly at github.com/Jannat332

Deterministic Traffic Light Controller `C++17` `CMake` `GoogleTest` github.com/Jannat332/traffic-light-controller ↗

Safety-Critical Embedded Finite State Machine Architecture

- Engineered a deterministic, cycle-accurate 4-phase intersection traffic controller in **C++17** enforcing strict safety invariants (preventing conflicting green lights under all conditions).
- Implemented asynchronous emergency vehicle preemption and pedestrian walk request handling with zero race conditions.
- Built an automated test suite achieving **100% test pass rate** across edge transitions, timeout triggers, and yellow-clearance timings.
- Structured with modular CMake configuration, RAII memory management, and zero raw pointers for high reliability.

LLM Parallel Reasoning & Verification Engine `Python 3.11` `RAG` `Docker` github.com/Jannat332/llm-parallel-reasoning ↗

Dual-Stream Semantic Retrieval & Knowledge-Graph Verification

- Architected a dual-stream reasoning pipeline combining dense vector retrieval (RAG) with relational knowledge-graph validation, **reducing hallucination rates by 42%** on multi-hop benchmarks.
- Designed a dynamic confidence-scoring fusion algorithm that detects and flags factual contradictions in LLM outputs before client delivery.
- Containerized the entire evaluation workflow using Docker, enabling reproducible benchmark execution with single-command deployment.

5G Network Slice QoS Verification Demo `Python` `3GPP Specs` `QoS/SLA` github.com/Jannat332/5g-network-slice-demo ↗

Telecom SLA Simulation: URLLC Mission-Critical vs. eMBB High-Throughput

- Built a Python-based 3GPP-aligned network slicing simulator modeling dynamic resource allocation between low-latency URLLC and high-bandwidth eMBB traffic.
- Implemented strict priority queuing and traffic shaping, ensuring **sub-1ms packet latency** for mission-critical industrial slices under heavy packet congestion.
- Engineered automated SLA verification and anomaly alerts when packet loss or jitter thresholds are violated.

Production CI/CD Matrix Pipeline Template `GitHub Actions` `Docker` github.com/Jannat332/github-actions-ci-template ↗ `DevOps`

Automated Multi-Platform Quality Assurance & Release Automation

- Constructed a reusable GitHub Actions CI/CD matrix pipeline executing cross-platform builds, automated linting, security vulnerability scans, and unit tests.
- Authored multi-stage Dockerfiles achieving a **65% image size reduction** via Alpine runtime base layers, non-root security boundaries, and layer caching.

PROFESSIONAL EXPERIENCE

Junior Software Engineer — Spectrum IT Solutions 2021 – 2022

Linux-Based Software Development & Backend Services

- Engineered, refactored, and maintained production software components and data endpoints using **C, C++, and Python** in Linux environments.
- Conducted rigorous root-cause defect analysis and memory optimization using **GDB and Valgrind**, resolving memory leaks and race conditions in mission-critical modules.
- Collaborated within an Agile / Scrum framework; authored automated unit tests, actively participated in peer code reviews, and managed version control through Git branches and PR workflows.
- Optimized backend database queries and caching layers, **improving data transaction response times by 25%**.

Research Assistant (Computational Systems) — North South University 2023 – 2024

High-Performance Computing, Numerical Modeling & GPU Acceleration

- Parallelized computational physics models using **C++, Python, and CUDA GPU kernels**, achieving up to **60% runtime acceleration** compared to serial baselines.
- Automated large-scale numerical data extraction, convergence validation, and statistical analysis pipelines.
- Co-authored 3 research papers published in peer-reviewed Elsevier scientific journals based on the computational findings.

EDUCATION & QUALIFICATIONS

Master of Science in Software Engineering and Digital Transformation — LUT University, Finland 2024 – 2025

CGPA: 4.24 / 5.00 (Distinction • Top 5% of Graduating Class)

Curriculum: Advanced Software Architecture, Distributed Systems, Cloud Computing, Quality Assurance, Agile Engineering.

Bachelor of Science in Computer Science and Engineering — North South University 2016 – 2021

CGPA: 3.44 / 4.00 • Thesis published in **IEEE Xplore** (Embedded sensor signal processing).

Aalto University F.E.C. (Further Educated with Companies) Program — Aalto University, Espoo, Finland Intake 2025 – 2026

Accepted participant. Provides host companies with a low-risk, subsidized 6-month recruitment traineeship transitioning into a permanent software engineering contract.

SELECTED SCIENTIFIC PUBLICATIONS

- **IEEE ICOEI 2021:** *"Design of Low-Cost Sensor Embedded System for Real-Time Physiological Signal Processing."* DOI: 10.1109/ICOEI51242.2021.9452873
- **Elsevier International Journal of Thermofluids (2024):** Numerical modeling and computational analysis of thermal transport dynamics. DOI: 10.1016/j.ijft.2024.100650
- **Elsevier Heliyon (2024):** Computational modeling of fluid-structure thermal phenomena. DOI: 10.1016/j.heliyon.2024.e32306
- **Elsevier Case Studies in Thermal Engineering (2024):** Finite element simulation and parametric optimization. DOI: 10.1016/j.csite.2024.104273